Impredicativity: Expressiveness

Dominik Kirst

MPRI PrfSys Course
October 23rd, 2026

Impredicativity and polymorphism

Recall the impredicative typing rule for Type:

$$\frac{\Gamma \vdash T : s \qquad \Gamma(T : s) \vdash U : \text{Type}}{\Gamma \vdash \Pi x : T.\, U \;:\; \text{Type}}$$

With impredicative rules for Type, we automatically get some polymorphic types:

$$\Pi T : \text{Type}.\, T \to T \;:\; \text{Type}$$
$$\Pi T : \text{Type}.\, T \;:\; \text{Type}$$

For completeness, a form of polymorphism for kinds would require a further rule:

$$\frac{\Gamma \vdash \alpha : \text{Kind} \quad \Gamma(T : \alpha) \vdash \beta : \text{Kind}}{\Gamma \vdash \Pi T : \alpha.\, \beta \;:\; \text{Kind}}$$

With that rule we would for instance have the polymorphic identity type:

$$\Pi X : \text{Type}.\, X \to X \to \text{Type} \;:\; \text{Kind}$$

Impredicative encodings

What are elements of the two impredicative types from last slide?

1. Easy to give an element of $\Pi T : \text{Type}.\ T \to T$, no extractable information
2. Hard to give an element of $\Pi T : \text{Type}.\ T$, could extract an element of any T

This essentially mimics the behaviour of the unit type $\top$ and the void type $\bot$!

There is a general pattern how (most) inductive types X can be encoded:

$$X' \ := \ \Pi T : \text{Type}.\, (\ldots \to T) \to \ldots \to (\ldots \to T) \to T$$

Here, we have one clause $(\ldots \to T)$ for every constructor with $\ldots$ as arguments.

We can easily define functions from X to X' and back:

1. Given $x : X$, $T : \text{Type}$ and functions $(\ldots \to T)$, build $t : T$ by recursion on x
2. Given $x' : X'$, the instance $x'\,X$ is an element of X after supplying the constructors

Round trip from X to X' and back is the identity, more subtle for the other direction.

Impredicative encoding of natural numbers

$$X' \;:=\; \Pi\, T : \mathrm{Type}.\, (\ldots \to T) \to \ldots \to (\ldots \to T) \to T$$

Applying the pattern to the type N of numbers with $O : N$ and $S : N \to N$ we obtain:

$$N' \;:=\; \Pi\, T : \mathrm{Type}.\, T \to (T \to T) \to T$$

The constructors and iterator are definable by

$$\begin{aligned} O' : N' \;&:=\; \lambda(T : \mathit{Type})(a : T)(f : T \to T).\, a \\ S' : N' \to N' \;&:=\; \lambda(n : N')(T : \mathit{Type})(a : T)(f : T \to T).\, f\,(n\,T\,a\,f) \\ R_T : N' \to T \to (T \to T) \to T \;&:=\; \lambda(n : N').\, n\,T \end{aligned}$$

and $R'_T\,(S'\,n)\,a\,f \rhd^* f\,(R'_T\,n\,a\,f)$ holds, but we cannot derive $O' \neq S'\,n$ or full induction.

Impredicative encodings of some simple types

Also other simple types we postulated can be encoded impredicatively

$$A \times B := \Pi T.\,(A \to B \to T) \to T \qquad A + B := \Pi T.\,(A \to T) \to (B \to T) \to T$$
$$\pi_1(t) := t\,A\,(\lambda ab.a) \qquad i(a) := \lambda Tfg.\,f\,a$$
$$\pi_2(t) := t\,B\,(\lambda ab.b) \qquad j(b) := \lambda Tfg.\,g\,b$$
$$(a,b) := \lambda Tf.\,f\,a\,b \qquad \delta_T(t,f,g) := t\,T\,f\,g$$

with the expected reduction rules following from simple function application:

$$\pi_1(a,b) \triangleright^* a \qquad \delta_T(i(a),f,g) \triangleright f\,a$$
$$\pi_2(a,b) \triangleright^* b \qquad \delta_T(j(a),f,g) \triangleright g\,a$$

Again, what we don't get is full dependent elimination, for instance needed for η-laws:

$$(\pi_1(t),\pi_2(t)) = t \qquad \delta_{A+B}(t,i,j) = t$$

Impredicative encodings of some dependent types

Even identity types and sigma types can be encoded

$$
\begin{array}{rcl@{\qquad}rcl}
I_T\,x\,y & := & \Pi P : X \to \mathsf{Type}.\, P\,x \to P\,y & \Sigma x : A.\, B\,x & := & \Pi T.\, (\Pi x.\, B\,x \to T) \to T \\
\mathsf{refl}_T\,x & := & \lambda P t.\, t & (a, b) & := & \lambda T f.\, f\,a\,b \\
L_P\,x\,y\,e & := & e\,P & \pi_1(t) & := & t\,A\,(\lambda a b.\, a)
\end{array}
$$

with the expected reductions $L_P\,x\,x\,(\mathsf{refl}_T\,x) \triangleright^* \lambda u : P\,x.\, u$ and $\pi_1(a, b) \triangleright^* a$.
However, by the lack of dependent elimination $\pi_2(t) : B\,(\pi_1(t))$ is not definable.

In summary, impredicativity bumps up the expressiveness of type theory:

- Polymorphism and impredicative encodings, especially of logical connectives
- Encodings with full dependent elimination are active field of research.
- Careless impredicativity opens the room for inconsistency...